The player

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Graal Analysis, revision 0.0.2.

This chapter is the body: how it moves, how it fights, and how it dies, from the first Graal in September 1998 through the later Windows builds. One era at a time. The step on the wire is the property message in chapter 1. The rectangle that replaces a cut bush is the partial-change message there. Neither is defined again here.

A tick, wherever a client was timed, is 50 milliseconds. Positions are in tiles unless a sentence says pixels, or unless the body is Graal3D's, which lives in its own world units. A build that was never opened for a rule is left unmeasured.

Across every 2D client that was actually timed, the walking step is half a tile each tick, ten tiles a second. What changes, by era, is the death clock, the sword's power, who puts a cut bush back, and how many tests a hurt has to pass before the hearts change.

September 1998: the first body

The first Graal is a Delphi client with a 50 ms frame timer. `[bin]` (`version-profiles/matrix/builds-static-survey.tsv`). Offline, under Wine, it walks 0.5 tile a tick: x went from 12.5 to 14.0 in three steps, 50 ms apart, 10.0 tiles a second. `[live]` (`version-profiles/matrix/live-runs-2026-10-30.md`). The length of that step is not an immediate in the walking routine. The live measurement is the number. The table inside the program was not read. `[bin]` (`version-profiles/matrix/rules-1st-graal.md`).

A candidate step is tested at six points. The test fails closed. A point is blocked when an object's box contains it, when it lies outside the 64 by 64 board, when either coordinate is below half a tile, or when the tile is not on the passable list. The list itself was not copied out. If any of the first four probes is blocked, the stepped axis is undone, the free axis is snapped, and a push counter increments. A reading above 37 is set back to 18. When the counter is 18 and a ledge-tile test passes, the player hops for 8 ticks along a facing table and then returns to the normal mode. A particular pattern of the six probes adds a nudge from a second table. Those offsets were not read. While pushing, once the counter is above 17 and the player is not carrying, the body steps through the push frames. `[bin]` (`version-profiles/matrix/rules-1st-graal.md`).

Fighting uses one hurt entry for a sword, a baddy touch, and the projectiles that were listed. A hit is refused while the action mode is already the hurt mode. An accepted hit drops whatever is being carried, sets that mode, sets a shared counter to 40, changes the hearts by the hit's power, and stores a knockback step. Hearts are a fractional count and do not rise above the integer heart cap. The counter then falls by one each tick. At 0 the mode returns to normal. While the counter, after that drop, still reads 39 down to 35, the player moves by the stored step: five steps. Twelve points are tested, four groups of three. A blocked step is undone for that tick. Walking is refused during hurt while the counter is still at least 30. Below that, until the counter hits 0, walking is allowed again and a new hit is still refused. `[bin]` (`version-profiles/matrix/rules-1st-graal.md`).

The drawing follows the same counter. Above 30 the hurt body is used. An even counter, or any counter still above 34, draws the full hurt frame list. An odd counter at or below 34 draws a single frame. Once the counter is 30 or below, the normal body is drawn, and odd counters again draw only one frame. `[bin]` (`version-profiles/matrix/rules-1st-graal.md`).

A baddy touch and the common projectile hit store an axis step of length 1 from the attacker's facing: facing 0 decreases Y, facing 1 decreases X, facing 2 increases Y, facing 3 increases X. A sword hit stores the delta between centres, victim minus attacker. The export then mishandles the length of that delta, the same class of bad assignment it shows elsewhere, so "away from the attacker" is the tagged direction and a unit length for a sword knockback is not established. There is no rule here that a zero step becomes one tile east. `[bin]` (`version-profiles/matrix/rules-1st-graal.md`).

The sword itself does not grow. Power is half a heart, fixed. Another player's swing is tested only while the local life-state is the normal value. The facing sits in the low two bits of the packed sprite and the frame sits above that. A hit is possible only on frames 6, 7, 8, 9 and 10. The box is measured between centres (each centre is x + 1.5, y + 2). The victim must sit ahead by a span of 4 tiles, and the sideways distance must be strictly under 2. On two facings the export includes both ends of that span and on the other two the ends are exclusive. That split is treated as decompiler noise. The tagged reach is "up to 4 tiles ahead, sideways strictly under 2", and the exact 0 and 4 edges are not tagged. `[bin]` (`version-profiles/matrix/rules-1st-graal.md`).

A baddy touch counts when both axes are within 1 tile of the body centres (player at x + 1.5, y + 2; baddy at x + 1, y + 1), and it takes half a heart. Two projectile types, sprites 65 and 261, take half a heart. The other types in that list take one heart. An arrow whose sprite is 107 to 110 is blocked, and does not hurt, when its direction is the opposite of the player's facing and the player's sprite index is below 6. Any other arrow takes half a heart. All of them use the same 40-tick hurt. `[bin]` (`version-profiles/matrix/rules-1st-graal.md`).

Death is immediate and local. When hearts fall to 0 or below, the heart write clamps them to 0, clears the shared counter and the action mode, sets the life-state to dead, and notifies the relay object when one exists. The payload of that notice was not read. There is no later status property on this path, and the inventory is not spilled. The dead life-state replaces the tile buffer with a filled rectangle and draws a text panel. The body uses one fixed sprite, index 157, only once the shared counter is above 27. Before that, with the action mode already cleared, the normal body is drawn. `[bin]` (`version-profiles/matrix/rules-1st-graal.md`).

A keypress while dead restarts the player once that counter is above 27. One other life-state, value 4, restarts with no counter test. The restart is the new-game setup: level `house1.graal`, position x 24.5 and y 30, facing 2 (increasing Y), hearts set to 3 and the heart cap set to 3 whatever the cap was before, carry and mode cleared. A second local player is shifted 12 tiles toward increasing X, to x 36.5 at the same y. The loop that advances the shared counter while dead was not found. Death itself sets the counter to 0, and both the dead sprite and the restart key wait for a reading above 27, so a 30-tick cap of the kind the later clients have is not tagged. No death sound is on this path. This build has no sound engine. `[bin]` (`version-profiles/matrix/rules-1st-graal.md`).

What that export still does not say: the six walk-probe offsets, the corner-nudge offsets, the numeric walk-step table, the exact 0-tile and 4-tile edges of the sword box, and the unit length of a sword knockback.

The year the step was timed and the rules were not

From the 10th Graal (1.0, December 1998) through 1.1 (January 1999), 1.32 (January 2000) and 1.41 (November 2000), the same offline timing gives the same step: 0.5 tile, 50 ms, 10.0 tiles a second. `[live]` (`version-profiles/matrix/live-runs-2026-10-30.md`). The static survey still records a 50 ms frame timer on each of those executables. `[bin]` (`version-profiles/matrix/builds-static-survey.tsv`).

1.1 starts the offline game with the player asleep in the left bed. A key wakes the player. `[live]` (`version-profiles/matrix/live-runs-2026-10-30.md`, `version-profiles/matrix/README.md`). After getting out of bed and bumping the furniture, x carried a 0.025 offset (26.925) and kept it while walking. A later wall rule

produces offsets of that kind, so a similar rule is likely. The 1.1 routine was not read. [live] (version-profiles/matrix/live-runs-2026-10-30.md).

1.32 was measured for that walking speed, on y, and nothing else in this chapter. A listing of the decompile folder on 2026-11-01, client exports and the accessories folder, still has no 1.32 export, so no death, hurt, sword or wall number was added for it. [bin] (version-profiles/matrix/rules-1st-graal.md). The rest of the builds between the first Graal and 1.39 — the 2nd through the 9th, the 1.1 gold builds, 1.21, 1.28, the 1.3 test, 1.31 — were not given this five-rule read either. Their frame timers are in the survey. Their hurt counters are not.

What did get read, from the 4th alpha through 1.41, is what a sword does to a bush, a pot and a stone, and who puts the object back.

The client rolls the drop. One Random(100): 0 to 24 a green rupee, 25 and 26 a blue rupee, 27 to 31 a heart, 32 to 36 bombs, 37 to 99 nothing. A sword cut of a bush or of tall grass rolls. Lifting a bush rolls. Lifting a pot always leaves a heart. A sign and a stone leave nothing. No client in this range rolls arrows. A bomb blast rolls for a bush and destroys a sign, a pot or a stone, and the pot still leaves a heart. [bin] (client-internals/item-drops.md). From the 4th alpha to 1.37 that roll goes to other players and not to the server. From 1.38 to 1.41 the client also tells the server. [bin] (client-internals/item-drops.md). The tell is the board-change message in chapter 1.

The board comes back from the server. On the 1999 server shipped with the 1.3 test, any change that is exactly 2 by 2 is restored after trunc(R · rand · 0.5 + 0.75 · R) one-second ticks, where R is respawntime and the default is 10, so 6 to 12 seconds. The 1.3 client and the 2.05 client keep no timer of their own. Offline, a cut bush stayed cut for the whole of a 633-second watch. [bin] [live] (client-internals/board-respawn.md). That 1999 server did not echo the change to the sender. The later lab that did echo, and waited 15.0 seconds, is the packet story in chapter 1.

February 2000: a body that is not on a tile grid

Graal3D is a separate program, dated between 1.32 and 1.33, with the same 50 ms timer. [bin] (version-profiles/matrix/builds-static-survey.tsv). It does not walk in tiles. The server beside it is in chapter 3. The body is here.

Forward, back, strafe and jump are the numpad. Walking speed is 8 world units a tick, 160 a second. After 25 ticks of walking without stopping, 1.25 seconds, the speed becomes 12 units a tick. Stopping resets the count. Holding only back walks backwards along the facing the player had when the key went down. A step that would leave the world, 4000 units either way on x and z, is dropped. While walking, the body bobs by a sine of the animation counter, using 3.14. [bin] (graal3d/client-behaviour.md).

A jump is a fixed table of upward speeds: 10, 10, 9, 8, 6, 3, 1, which is 47 units in 7 ticks. The same table is then walked back down, and after that the body falls at 10 units a tick until it lands. Letting go of the key early starts the fall at once, so a tap is a short hop. A new jump needs the key released and pressed again. [bin] (graal3d/client-behaviour.md).

There are no chat bubbles. A line joins a log at the bottom left, white Dialog text, and leaves after ten seconds. A line longer than two hundred characters is cut. [bin] (graal3d/client-behaviour.md).

The follow camera sits about 220 units behind the player and about 80 above, turns at most five degrees a tick, and then pitches down by a fraction of a radian. [bin] (graal3d/camera.md). Numpad 0 toggles follow. The field of view is the library's initial field, a square window, an opening of about 53.13 degrees on either axis. The executable does not set it. [doc] [bin] The far plane was not read back out of this executable. The library's

initial back plane and a later DelphiX fork give two different numbers, and neither is tagged as this client's far plane (`graal3d/camera.md`).

Collision tests the new position against every face of the level mesh. The collider's radius is 5. A walkable ledge up to 30 units above the feet can be stepped onto. Steeper faces than an upward normal of 0.2 still snap the point and skip the mesh attach. A probe that reports blocked stops the rest of the probes for that step. `[bin]` (`graal3d/collision.md`, `graal3d/client-behaviour.md`). This path was not confirmed in the lab.

The behaviour note records walking, running, jumping, the camera and the chat log. It records no sword, no heart count and no death sequence. A Wine lab got the client through login and as far as saving the level file. It could not show the world, and it could not be walked around. `[lab]` (`graal3d/lab-2026-10-30.md`). Combat and death for this program were not measured.

August 2000: 1.39, the last client before ganis

1.39 is the build where the five rules were read again, and then watched. The frame timer is still 50 ms. `[bin]` (`version-profiles/matrix/builds-static-survey.tsv`). The walk is still 0.5 tile a tick when unburdened, and 0.2 of that when hidden. Horse and shield have other multipliers. Those multipliers were not written down as numbers. `[bin]` (`version-profiles/matrix/rules-139.md`).

The wall test is six probe offsets per facing step. A failed probe refuses the step. The blocked-step counter still pushes at 18 and wraps from 38 back to 18, and that counter was seen live as well as read. `[bin]` `[lab]` (`version-profiles/matrix/rules-139.md`). Sliding along a blocking character nudged the player by 0.125 tile a tick. That slide is a consequence of the probe path. A touch script did not fire on that contact. `[lab]` (`client-internals/player-mannerisms-live.md`).

The sword reach is the same shape as the first Graal, now with both ends tagged: 0 to 4 tiles ahead of the attacker's centre, sideways strictly under 2. `[bin]` (`version-profiles/matrix/rules-139.md`). Offline, the sword key runs a counter from 1 to 6, five ticks, and cuts the bush in front on the first frames. `[lab]` (`client-internals/player-mannerisms-live.md`). There is still no gani. The animation is built into the executable. The hurt pose shows the player's back with the arms out for about 10 ticks, and only then does the body blink, every other tick, out to tick 40. `[lab]` (`client-internals/player-mannerisms-live.md`).

A hit is refused when the player is already in hurt mode, in mode 7, or outside the playing life-state. An accepted hit clears the current pose, sets hurt mode and a counter of 40, applies the power (a negative power heals, and hearts stop at the maximum), and stores a knockback unit vector. A zero vector becomes one tile east. If something was being carried, a positive power may force a drop. Each tick the counter drops by one. While it is strictly greater than 34 after that drop, readings 39 down to 35, the player moves one tile along the stored vector. Twelve body-box points are tested, and a blocked step is undone. `[bin]` (`version-profiles/matrix/rules-139.md`).

The offline touch test measured four tiles of knockback. The counter window in the program is five steps, the same window as 2.05. Both figures are quoted in chapter 4. The note that records them offers a blocked step, or a one-tick error in the video, as a way one step could be hidden. That offer is not a measurement that the fifth step happened.

A baddy touch takes 0.5 hearts, or 1.0 for type 7, when the player is within 1 tile on both axes of the baddy centre. `[bin]` (`version-profiles/matrix/rules-139.md`).

Death now has a clock. Hearts at or below 0 are clamped to 0, the hurt counter is cleared, the mode returns to normal, and the life-state becomes dead. `dead.wav` plays. The inventory is partly spilled as floor items, rupees in

bands of 100, 30, 5 and 1, and arrows and bombs similarly. Each tick while dead, a counter under 30 increments, so it stops at 30. Between 4 and 27 the client cycles death sprites. A movement or action key respawns once the counter is above 27. The dead bit of the status property is included only when the life-state is dead and the counter is above 27. [bin] (version-profiles/matrix/rules-139.md).

Respawn clears the hurt counter and the modes, restores hearts to the maximum through the same write used when filling hearts, and warps to the last saved level entry when one exists. Online, with no saved name, the place is x 30.5, y 30.0 in the current level. Offline, with no saved entry, the built-in start-level table is used. There is no respawn packet on this path. The heart restore has no alignment-band test. 1.39 has no alignment bar. The notes place those bands at 1.41. [bin] (version-profiles/matrix/rules-139.md, version-profiles/matrix/README.md).

Swimming starts when x + 1.5 lies on a liquid tile, at the same 0.5 tile a tick. Standing still, only the head and a water ring are drawn, and the ring keeps alternating. [lab] (client-internals/player-mannerisms-live.md). Hiding under a carried bush sets a hidden flag and walks at 0.1 tile a tick. The sword key throws from hiding. [lab] (client-internals/player-mannerisms-live.md). The exact numeric contents of the six walk probes and the twelve knockback probes were not copied out.

The builds from 1.33 to 1.38 were not given this rule read. What the version note records for them is the wire: login, which ids they handle, and, at 1.38, a change in how a weapon is registered. [bin] (client-internals/versions-133-139.md). Their step, their sword box and their death clock were not measured one by one. 1.39 is the measured end of that row.

November 2000, then January 2001: the alignment bar, then the first ganis

1.41 rev 1 still has the 50 ms timer, and the offline walk is still 0.5 tile a tick. [bin] [live] (version-profiles/matrix/builds-static-survey.tsv, version-profiles/matrix/live-runs-2026-10-30.md). The in-game picture adds the alignment bar under the item buttons, and a small number beside the sword. [live] (version-profiles/matrix/live-runs-2026-10-30.md). The death routine of this 1.41 was not read, so the 1.39 death clock is not assigned to it.

NewMain 1.41.1, the January 2001 update pack graalnm24, is the first gani client in the mannerisms run. It was traced the same way as 2.05. The walk is 0.5 tile a tick over an 8-tick cycle. The push counter is the same 18, wrapping 38 back to 18, and each push frame lasts 4 ticks. A lift plays for 6 ticks and then a carried-still pose. The sword plays for 5 ticks and cuts the bush on the first frame. Hurt refuses a second hit for 40 ticks, knocks the player back 5 tiles, holds the hurt pose for 31 ticks, and blinks from the hit itself, every other tick. [live] (client-internals/player-mannerisms-live.md).

Two habits are this build's own. Hiding sets the hidden flag and slows the walk to 0.1 tile a tick, and the pose stays the carry pose. The gani folder does contain hidden poses, and this client does not play them. The legs keep cycling, one frame a tick, at a fifth of the speed. Letting go of the arrow during a pull returns to the grab pose. Standing still in water resets the swim frame to 0 and freezes it. [live] (client-internals/player-mannerisms-live.md).

The server takes the drops, and the second local player leaves

NewMain 1.42 is where the release notes move bush rupees and pot hearts onto the server. The roll is gone from NewMain 24 and 26. The 1.41 program graal_gr1411 still has it. [doc] [bin] (client-internals/item-drops.md). The feature list places the same sentence on 2.00, together with the removal of two-player, three-player and four-player local play, and of the save-game keys. [doc] (client-internals/feature-

`timeline.md`). The matrix dates the loss of local multiplayer to NewMain 1.42 and to 2.00. `[bin]` `[live]` (`version-profiles/matrix/README.md`).

From NewMain 1.42 onward the surveyed executables have no frame timer. The frame runs from the idle handler. `[bin]` (`version-profiles/matrix/builds-static-survey.tsv`, `version-profiles/matrix/live-runs-2026-10-30.md`). That is a change in how the program wakes. It is not a measured change in the step. 2.00 itself was not timed. Its training level opens with a scripted scroll that holds the player, and the scroll did not finish inside the run. `[live]` (`version-profiles/matrix/live-runs-2026-10-30.md`).

The sword judge was read, with no wire and no lab, in 2.00, 2.01, 2.02, 2.03 and 2.04. 2.00 revisions 1 to 3, and 2.02 revisions 1 and 2, were not read. None of these five is without combat. In every one of them the victim's client judges another player's swing from that player's position, direction and sprite. The swing counts on animation steps 9 to 13, which is sword frames 6 to 10. The victim must be 0 to 4 tiles ahead, both ends included, and strictly under 2 tiles to the side, measured between centres with the same 1.5 and 2 offset. The box test is skipped while the victim is already in hurt mode. Damage is the attacker's sword power, as last received, in half hearts. Hurt mode lasts 40 ticks, and the knockback vector is normalised. A heal needs alignment at or above the same threshold later read as 40. The hurt packet takes a power capped at one heart, from an attacker within 10 tiles. `[bin]` (`client-internals/combat-200-204.md`).

2.00 is the oldest of those shapes. It has no test that refuses a hit while the attacker plays a kick, no shared "can be hurt" test inside the sword judge, no script event on the hit, and no second hurt packet. Its own victim test requires the play state to be exactly the normal one, so a player who is in some other state is not hit by a sword. Sword power is stored in 6 bits, 0 to 63, and a negative power does not survive that store. The once-a-second limit and the alignment-100 immunity are absent from its hurt packet. They are present from 2.01. The kick test and the hit script event are present from 2.01. From 2.03 the stored power is limited to the range −20 to 20. 2.04 matches 2.05 revision 2 on the sword judge, the reach box, the hurt routine, the heart change and both hurt packets. The no-kill flag, which 2.00 through 2.03 keep on the player and clear when a level loads, sits on the level in 2.04. `[bin]` (`client-internals/combat-200-204.md`).

That read was not tested in the lab or on a wire. The two attacker flags are inferred to be the same status bits 2.05 tests. Where the decompile keeps the call arguments, in 2.00 and 2.02, the sword knockback handed to the hurt routine is the victim's centre minus the attacker's point, halved, and the hurt routine then normalises it. In the other builds of this row the decompile drops those arguments, so they were not compared. `[bin]` (`client-internals/combat-200-204.md`).

2.05: the player who was taken apart

2.05 revision 2 is the client whose walking, sword, hurt and death were all read, and then watched tick by tick. Its form has no timer. The live step is still 0.5 tile every 50 ms. `[live]` (`version-profiles/matrix/live-runs-2026-10-30.md`). The same half-tile vectors, (0, −0.5), (−0.5, 0), (0, 0.5) and (0.5, 0), are in the walking routine of 2.05, 2.17.1, 2.22 and 2.31, multiplied by a speed factor that is 1.0 on foot. `[bin]` (`client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md`). A high-speed state uses 1.2. Swimming and carrying have other factors. Those were not read in the first pass over the body box. `[bin]` (`client-internals/decomp-findings-2026-10-25-l/01-walking-body-box.md`). Hidden walking, read later and seen live, uses 0.2, which is 0.1 tile a tick. `[bin]` `[live]` (`client-internals/carry-hide-and-glove.md`).

One key is one step, in the order up, left, down, right. A diagonal is two steps, each tested at the position the previous one left. The first act of a step is to face that way. Six points lie on the leading edge of a box two tiles square, from x + 0.5 to x + 2.5 and from y + 1 to y + 3, evaluated where the step would land. The first four decide

the step. If all four are free, the step is kept. If any is blocked, it is refused and the player is put in the pushing state. [bin] (client-internals/decomp-findings-2026-10-25-l/01-walking-body-box.md, client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md).

A refused step can still move the body. If the corner pattern matches, 0.1 tile is added across the edge, the step itself is taken back, and the facing coordinate is snapped so the body's edge ends on a tile line. There is no wall test on the snapped position. Pressing against the end of a wall therefore slides the body around it at 0.1 tile a tick, ten ticks to clear one tile. Moving left or up, the probe is already inside the body, so the edge meets the wall without that snap. Moving right or down, the probe is the tile line one tile beyond the body, and without the snap the body would stop half a tile short. A two-tile-wide body fits a two-tile gap and does not fit a one-tile gap. [bin] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md).

A point is blocked, in order, by a flagged object list, by a blocking NPC, by a list of 1.5 by 2 boxes, by a chest (2 by 2, edges included, opened or not), and then by a tile that is in neither passable list. Lava is a liquid and is walkable at this test. The first object list was not identified. A coordinate becomes a tile as the floor of the value plus a ten-thousandth, so a point exactly on a line reads the tile that starts there. [bin] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md). The passable lists this routine consults, and the other counts in the format notes, are the collision disagreement in chapter 4.

Live, the first tick after the key goes down already moves the player, and one gani frame advances with it, eight frames and then back to the first. Letting go switches to idle in that tick. There is no slow-down and no finishing step. Against a wall the walk frame keeps cycling in place, and the push pose replaces it on the 18th blocked tick, 0.9 seconds. The push pose is five frames of four ticks, and it restarts when the counter wraps. Holding a second direction along the wall walks away at full speed in the same tick, and the blocked count drops, so the 18 ticks have to be earned again. [live] (client-internals/player-mannerisms-live.md). The same 18th refused step, at a ledge tile for that facing, starts an 8-tick hop. The walk pose stays up. Keys are not read until the landing. The live y values of an upward hop match the curve in the program. [bin] [live] (client-internals/ledges-push-pull.md, client-internals/live-trace-205.md).

The action key, in order, throws what is carried, lifts, or grabs. A lift lasts 6 ticks. The object counts as carried from the 5th, and the carried-still pose follows. Walking while carrying is still 0.5 tile a tick. Players start at glove power 1, which lifts a bush, a sign and a pot. The next glove adds the light stone, and the one after that the black stone. The sword key throws from hiding, and the program also throws on the sword key while merely carrying. That second case was read and not traced live in 2.05. [bin] [live] (client-internals/player-mannerisms-live.md, client-internals/carry-hide-and-glove.md). Grabbing a wall holds for as long as the action key is held. The arrow away from the wall plays a pull, and a pull is tried every 18 ticks. Letting go of the arrow leaves the pull pose looping until something else moves the player. A pull against a plain wall moves nothing, and the trigger still fires. [live] (client-internals/player-mannerisms-live.md).

A sword swing needs weapons allowed, a sword power other than 0, and a player who is not swimming without a horse and not already spinning. A normal swing is one mode. A spin is another: it needs the has-spin status, and it starts when the sword key is pressed again while the swing is still running. It is not a hold to charge. Holding the key for 2.5 seconds gave no charge, with or without the spin status. The swing is five frames of one tick, and the player does not move during it. The bush in front is cut on the first frame. A second press on frame 2, 3 or 4 cuts the swing off and plays a 13-tick spin. A press after the swing has ended is a new swing. 2.22, online, did the same. [bin] [live] (client-internals/player-mannerisms-live.md, client-internals/decomp-findings-2026-10-24/04-sword-hitbox.md). Which bushes a spin cuts, and whether a spin hits a player, were not checked.

The victim decides the hit. The attacker broadcasts position, direction and sprite, which is the walk message. A hit counts when the attacker is on the hitting frames, the victim is 0 to 4 tiles ahead and strictly under 2 to the side,

the victim's alignment is not exactly 100, and the victim is hurtable: not already hurt, not swimming, not dead, not in the editor. [bin] (client-internals/decomp-findings-2026-10-24/04-sword-hitbox.md). Half-hearts taken equal the attacker's sword power, limited to −20 to 20. In hearts that is the power divided by 2, so a power-3 sword takes 1.5 hearts. A negative power heals only at alignment 40 or above, and hearts stop at the maximum. Hurt mode starts even for a power of 0 and even for a heal that was ignored. [bin] (client-internals/decomp-findings-2026-10-24-d/01-sword-damage.md). The victim's shield, the item levels and the attacker's alignment were not read into that formula.

The hurt that follows is mode 3 for 40 ticks. The knockback vector is scaled to one tile, and a zero vector becomes one tile east. For the five ticks while the counter reads 39 down to 35, the player moves one tile along it, then twelve points on a box centred at x + 1.5, y + 2 are tested. A blocked step is undone and tried again next tick. On open ground the player ends five tiles away after 250 ms. The same routine and the same twelve points are in 2.05 revision 2, 2.17.1, 2.22 and 2.31. [bin] (client-internals/decomp-findings-2026-10-25-k/01-knockback-and-packet-hits.md). Live, a touch that takes half a heart (3 hearts becoming 2.5) moved the player 5.0 tiles, from (29.5, 48) to (28.79, 52.95). Inside a large blocking image every step was refused and the player did not move. The hurt pose lasts 31 ticks when no arrow is held, and a held arrow walks again from the 12th tick. Blinking, body present and then absent, runs the whole 40 ticks, including after the pose has gone back to idle. [live] (client-internals/player-mannerisms-live.md).

Other hurts call that same routine, so a second piece of one blast does not take hearts again while the mode lasts. An ordinary bomb piece and a super-bomb piece take 0.5 hearts. A jolt-bomb piece takes 1. An arrow takes 0.5. A fireball takes 1. A fireblast takes 0.5 and then a blast. A nuke takes 1 and then a blast. A thrown object takes 1, or 0.5 when the player has exactly 1 heart left, if it lands within 2 tiles of (x + 0.5, y + 1) on both axes. A baddy takes 0.5, or 1 for the golden warrior, when the player's centre is within one tile of the baddy's point on both axes. A bomb, blast, arrow or throw that belongs to another player does not hurt after the level has run the no-kill command. A source that belongs to nobody, a baddy or the level itself, always hurts. When a player-owned source leaves the hearts at 0 or below, the client names that owner as the killer. [bin] (client-internals/damage-sources.md).

A bomb piece hurts when its centre, offset by one tile, is less than one tile from (x + 1.5, y + 2) on both axes. The live capture of the player's own bomb took the reported hearts from 6 half-hearts to 5 in the same tick, with the hurt pose, and the server had not asked for that report. [bin] [cap] (client-internals/damage-sources.md, client-internals/live-trace-205.md). A direct hurt packet carries its own power, capped at one heart, and is refused beyond 10 tiles. A power of 0 still starts hurt mode and knockback. Those gates, including the 10-tile test, were confirmed on the wire with 2.22. [bin] [cap] (client-internals/decomp-findings-2026-10-25-k/01-knockback-and-packet-hits.md, client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md). The 2.22 capture is more specific about the power byte and the distance, and it is quoted with that client below.

Swimming is the same 0.5 tile a tick, from the tick the point (x + 1.5, y + 2) is on water, and the stroke keeps animating while the player stands still. [live] (client-internals/player-mannerisms-live.md). Lava is two different tiles in the live level. The bright-red tile swims and burns. The dark-red floor stays a walk and burns. On either, a counter runs 0 to 100 and wraps, 101 ticks, and each wrap costs half a heart. Three wraps took 3 hearts down to 1.5. Stepping onto grass resets the counter. Burning to 0 offline shows the end scroll. [live] (client-internals/live-trace-205.md). A packet capture of that burn was not in the hearts-report list at the top of the same note.

Death, in the program, starts when any change leaves the hearts at 0 or below, and a per-frame check forces the dead life-state whenever they are already there. A timer starts at 0 and stops at 30, which is 1.5 seconds. The mode returns to normal and the player faces down. The local dead pose plays, and a script event fires. Controls are

ignored until the timer has passed 27. The dead bit of the status property is set only when the life-state is dead and the timer is at least 28, so other players see the death about 1.4 seconds late. No dead sprite goes on the wire. The sprite property stays idle plus direction. [bin] (client-internals/decomp-findings-2026-10-24/01-death-respawn.md).

There is no respawn packet and no server permission on this path. After the timer has passed 27, any movement or action key respawns. Hearts come back by alignment: below 20 gives 3, 20 to 39 gives the minimum of 5 and the maximum, 40 or more gives the maximum. The destination is the last saved level entry, updated on level entry unless the old and new levels share a name group, and also updated at login. The exact "same name group" rule was only partly traced. With nothing saved, online, the place is x 30.5, y 30.0, facing down, in the current level. Offline it is the tutorial level tutorial_b-4.graal. Online, the client re-sends the current level at the current position, so the server sees an ordinary level warp. [bin] (client-internals/decomp-findings-2026-10-24/01-death-respawn.md).

On a loopback stub that sent hearts of 0 and did not send alignment, the real 2.05 client answered at once with the dead pose, and about 1.35 seconds later with a status that includes the dead bit. A key then warped to the login point of that session, (30, 30), in the same frame as hearts of 3 (reported as 6 half-hearts) and a cleared dead bit. It did not wait for a reply. Sending a maximum of 0 before the kill did not stop that report of 3 hearts. [live] (client-internals/live-trace-205.md). The program's unsaved online place is x 30.5, y 30.0. The live trace's place is the login point of that session, (30, 30). They are two observations. This chapter does not average them.

Hearts, alignment and maximum power sit in a store the client obfuscates twice and cross-checks. A mismatch raises a memory-problems error. A scan from outside the process does not find the hearts as a plain value. [bin] [live] (client-internals/decomp-findings-2026-10-24/01-death-respawn.md, client-internals/live-trace-205.md).

The sword still does not regrow a bush. Cut and lift write the leftover tiles and, only when online, tell the server, using the board-change message in chapter 1. The 2.05 client spawns no rupee and no heart of its own. Offline, nothing comes back. [bin] [live] (client-internals/board-respawn.md, client-internals/item-drops.md).

The same step, as far as 2.31

2.17.1, 2.22 and 2.31 share the walking tables, the corner nudge, the flush snap, the knockback box and the sword-power formula with 2.05. [bin] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md, client-internals/decomp-findings-2026-10-25-k/01-knockback-and-packet-hits.md, client-internals/decomp-findings-2026-10-24-d/01-sword-damage.md). The ledge counts, the flight curve, the push and pull blocks and the pull cycle are the same in 2.22 and 2.31. Those two grab any blocking NPC, where 2.05 grabs a blocking NPC only when a per-NPC flag is clear, and their NPC lookup also searches the neighbouring parts of a map. From 2.22, walking, the ledge test and the action key do nothing while either of two server-set stop flags is on. [bin] (client-internals/ledges-push-pull.md).

Live, 2.22 matches 2.05 on lift, hide, the stuck pull, push, swimming in place, and the hurt: five tiles of knockback, and 12 ticks before a held arrow walks again. The one difference seen is that the swim frame restarts at 0 when the player stops, and then runs on. The sword key while carrying, and not hiding, goes to idle in the next tick with no sword frame: it throws. 2.31 was not run for these mannerisms. [live] (client-internals/player-mannerisms-live.md).

The 2.22 lab is where the death arithmetic was seen on the wire. The clock itself is the 2.05 program above. After the death key the client reports hearts by the alignment brackets: alignment 10 and 19 give 3 hearts, 20 and 39 give 5, and 40, 60 and 100 give the full maximum that had been set. The warp request is the level's entry point, (30.0, 30.5) on that level, and a second death during a delayed reply asks for the same entry, not the spot where the player fell. `[cap]` (`client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md`). That entry point is a third place, beside the 2.05 program's (30.5, 30.0) and the stub session's (30, 30). Each keeps the condition the note gave it.

A hurt packet at a squared distance of 100 lands, and at 110.25 it does not. A power byte of 0 knocks back and takes no hearts. A power of 1 takes half a heart. A power of 2 or more, up through the values that were sent, takes one heart. `[cap]` (`client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md`).

2.12, 2.13, 2.14, 2.15 and 2.16 have no row in the executable survey and no release between the 2.1 test builds and 2.17 in the client archive. Their walk, their sword and their death were not measured. 2.3 has no executable in the notes. The neighbour sizes and the neighbour rules are not assigned to it. `[bin]` (`version-profiles/matrix/README.md`, `version-profiles/2.22-2.31.md`).

From 2.3 the release notes describe full-precision positions, enabled once every client has switched, and the classic client-side mode as removed. `[doc]` (`client-internals/feature-timeline.md`). 2.31 handles the precise-move packet. That a server of the 2.3 era already sent it is not shown by the client's receive switch. `[bin]` `[inf]` (`client-internals/feature-timeline.md`).

After the Delphi client

The 3.x, 4.x, 5.x and 6.x programs are a different engine. The measured ones still take a step of half a tile, and they still let the server put a bush back. The death sequence itself — the timer, the dead bit, the alignment bands, and the place a key returns you to — was not measured on any of them. Those rules stop at 2.22's wire and at the 2.05 program.

The Linux 3.04 client, and a Mac 3.153

No Windows 3.x client was run. The 3.04 that was played is the Linux build. It walks 0.5 tile every 50 ms, ten tiles a second. Against a wall the push pose comes after about 0.9 seconds. A sword on a bush lasts 0.25 seconds and then idle, and the client sends the board change. Lifting a bush becomes a carried-still pose after 0.3 seconds. Throwing a bush sends no throw packet. While moving, position updates go to other players every 100 ms, and once more on stopping. Idle follows the key release by about 0.1 to 0.15 seconds. `[cap]` (`version-profiles/lab-304-linux-2026-10-30.md`, `version-profiles/3.04-3.153.md`).

Online, the bush comes back when the lab server says so, 14.6 seconds in that run, which is that server's setting. Offline, classic bushes, stones and pots draw wrongly, the player walks through them, and the sword does not cut them. There is no offline respawn to measure, because the objects do not break. `[cap]` (`version-profiles/lab-304-linux-2026-10-30.md`). Hurt, knockback and death were not part of that session.

The Mac 3.153 client, on the same kind of lab, sends the precise coordinates and never the old ones. In that session the moving axis went out about every 150 ms. The size of one step was not measured, so the half-tile rule is not assigned to it from this run. A sword was followed by idle 0.75 seconds later. That gap was not counted in ticks, and the session was also fetching the sword pose and the sword sound, so 0.75 seconds is not a swing length. `[cap]` (`version-profiles/lab-mac-tiger-2026-10-31.md`).

Windows, from 4.04

The Windows 4.04 revision 2 client was run under Wine against a loopback lab server. While a key is held it sends a precise coordinate every 50 ms. A step is 8 pixels, which is half a tile, ten tiles a second. Idle goes out one tick after the last step. Against a wall the push pose starts about 0.9 seconds after the player stops, and a push trigger goes out about 0.55 seconds after that pose starts, then about once a second. [live] (client-internals/live-behaviour-404.md).

The sword returns to idle about 250 ms later. A lift reaches the carried-still pose about 300 ms later. Hiding under a bush, a stone or a black stone draws only the object and walks at 1.6 pixels a tick on average, 0.2 of a normal step. The action key on a pot does not hide. The sword key throws from hiding. With the starting glove, the action key on a stone grabs. With glove power 3 the black stone lifts. [live] (client-internals/live-behaviour-404.md).

The bush, the pot and the stones come back only when the server sends the old tiles. In this lab that was 14.2 to 15.0 seconds, the lab server's 15-second timer checked once a second. The client keeps no timer. How long an original server of this era waited was not measured here. [live] (client-internals/live-behaviour-404.md). The player's own death, hearts to zero and the return point, was not measured.

A restored 4.20 revision 8 engine, the packet-logger build with its hook put back, walked at the same 8 pixels a tick. Its push pose started 0.85 seconds after the last step, and the push trigger followed on the same schedule. Carrying and hiding were not repeated on it. The stock 4.20 revision 8 archive could not be opened. [live] (client-internals/live-behaviour-404.md, protocol/v4x-clients.md).

Logging in to a server whose name begins with "login" draws no player and no hearts. The level is a backdrop for the server list. A warp message closes that session and opens the game server, and then the body, the hearts and the item buttons appear. Walking, chat and the sword are sent during the login session. Only the drawing is cut. [live] [bin] (client-internals/live-behaviour-404.md).

Windows 5 and 6

5.24 revision 4, 5.30 revision 3 and 6.03 revision 7 were played on a lab server and timed from their own messages. They agree with each other, and with 2.05, on the step: 8 pixels, half a tile, every 50 ms. Push starts 0.85 to 0.90 seconds after the walk into the wall. While pushing, a push trigger goes out about once a second. A lift reaches carried-still 0.30 seconds later. A throw is idle in the same tick, and this time a throw packet is sent. The sword and the bush cut are in the tick the swing starts, and idle is 0.25 seconds later, five ticks. [wire] (protocol/v5v6-lab-play.md).

Hiding under a bush, a light stone or a black stone sets the hidden bit, draws only the object, and moves about 1.5 pixels a tick, reported as about 0.1 tile a tick, the 2.05 rate. The action key on a pot does nothing toward hiding. Grabbing starts in the next tick. A pull against a stone that the glove cannot lift does not move the player or the stone, and a pull trigger goes out 0.85 seconds into the pull. [wire] (protocol/v5v6-lab-play.md).

An explosion on the player plays hurt and changes the hearts in the same tick, 3 to 2.5 for the power that was sent. The player is pushed 5 tiles in 5 ticks, one tile a tick, and blinks. Idle comes 1.54 to 1.55 seconds after the hit, which is the 31 ticks of the hurt pose. Swimming, into the deep-water tile used in that level, moves at the walking 0.5 tile a tick. Only the head is drawn. Stepping out returns to the walk. [wire] (protocol/v5v6-lab-play.md).

A hurt packet sent straight at 5.24 revision 4 changed nothing, whether the attacker id was 0 or a player that had been added. Only the explosion hurt the player. Why that packet was ignored was not read from the program.

`[wire]` (`protocol/v5v6-lab-play.md`). The 2.22 client, on the earlier lab, took that packet inside 10 tiles. `[cap]` (`client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md`). The 5.24 result is one lab send. The packet was not retried on the other 5.x builds, and the program was not read for a reason.

An earlier pass of the same era timed 5.00 revision 7, 5.12 and 5.24 revision 4 from their messages. All three walk 0.5 tile, 8 pixels, a tick. Idle is the next tick after the key comes up. Push follows the last refused step by about 900 ms, 18 ticks. The sword returns to idle in about 250 ms, five ticks. 5.00 and 5.24 reach a carried-still pose about 300 ms after a lift, six ticks. Hidden walking was measured on 5.24 only: 35 pixels in 22 ticks, 0.1 tile a tick. 5.12's lift was not run, because the bush was already cut. Against a corner, 5.00 nudged 1 or 2 pixels a tick toward the gap. `[cap]` (`version-profiles/lab-v5-2026-10-30.md`). 5.31's step, sword and hurt were not measured. What that build showed, before and after the warp, is which tiles it draws (`protocol/v5v6-lab-play.md`).

The bush comes back when the lab server sends it, 14.6 to 14.7 seconds in the 5.24 and 6.03 runs. These clients have no tile timer of their own. The wait an original 5.x or 6.x server used was not observed. `[wire]` `[srv]` (`protocol/v5v6-lab-play.md`).

6.00 revisions 2 and 4 freeze under Wine after the first image. On a Windows machine they did not. Arrow keys walked the player, and a baddy's hit took the hearts from 3 to 2.5. The step size, the knockback distance and the hurt length were not measured in that run. `[shot]` (`protocol/v5v6-lab-play.md`). Future runs of that kind stay on loopback. That one did not, and the note says so.

The same login-server rule as 4.04 hides the body until a warp. Before the warp, 5.31 and 6.03 draw chests and image NPCs and no tiles. After it, with or without an extra tile-definition script, they draw the level. `[wire]` `[shot]` (`protocol/v5v6-lab-play.md`).

Hearts reaching 0, the death timer, the dead bit, the alignment bands and the respawn point were not measured on 4.04, 4.20, 5.24, 5.30, 6.00 or 6.03. The hurt those labs did record stops at half a heart and a five-tile push.